例设 $u=f(x,y,z), y=\varphi(x,t), t=\psi(x,z)$，其中 f,φ,ψ 均有连续的偏导数，求 $\dfrac{\partial u}{\partial x}$ 和 $\dfrac{\partial u}{\partial z}$.

解法一：利用全微分形式的不变性

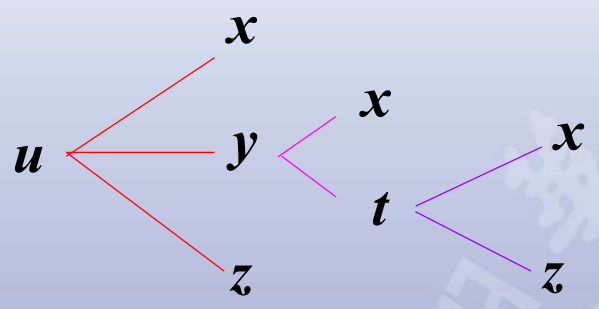
$$
\begin{aligned}
du &= f_x dx + f_y dy + f_z dz \\
&= f_x dx + f_y\left(\varphi_x dx + \varphi_t dt\right) + f_z dz \\
&= f_x dx + f_y\left[\varphi_x dx + \varphi_t\left(\psi_x dx + \psi_z dz\right)\right] + f_z dz \\
&= \left(f_x + f_y\varphi_x + f_y\varphi_t\psi_x\right)dx + \left(f_y\varphi_t\psi_z + f_z\right)dz
\end{aligned}
$$

所以 $\dfrac{\partial u}{\partial x} = f_x + f_y\varphi_x + f_y\varphi_t\psi_x \qquad \dfrac{\partial u}{\partial z} = f_y\varphi_t\psi_z + f_z$

例设 $u=f(x,y,z), y=\varphi(x,t), t=\psi(x,z)$，其中 f,φ,ψ 均有连续的偏导数，求 $\dfrac{\partial u}{\partial x}$ 和 $\dfrac{\partial u}{\partial z}$.

解法二：利用复合函数求导法

这里函数复合关系为



$$\frac{\partial u}{\partial x}=f_x+f_y\frac{\partial y}{\partial x}+f_y\frac{\partial y}{\partial t}\frac{\partial t}{\partial x}$$

$$=f_x+f_y\frac{\partial \varphi}{\partial x}+f_y\frac{\partial \varphi}{\partial t}\frac{\partial \psi}{\partial x}$$

$$=f_x+f_y\varphi_x+f_y\varphi_t\psi_x$$

$$\frac{\partial u}{\partial z}=f_y\frac{\partial y}{\partial t}\frac{\partial t}{\partial z}+f_z=f_y\varphi_t\psi_z+f_z$$

解法三：视为方程组 $\begin{cases} u = f(x,y,z), \\ y = \varphi(x,t), \\ t = \psi(x,z). \end{cases}$ 确定的隐函数

$u = u(x,z), y = y(x,z), t = t(x,z)$ 的偏导数问题.

方程组 $\begin{cases} u = f(x,y,z) \\ y = \varphi(x,t) \\ t = \psi(x,z) \end{cases}$ 两边对x求导得 $\begin{cases} \dfrac{\partial u}{\partial x} = f_x + f_y \dfrac{\partial y}{\partial x} \\ \dfrac{\partial y}{\partial x} = \varphi_x + \varphi_t \dfrac{\partial t}{\partial x} \\ \dfrac{\partial t}{\partial x} = \psi_x \end{cases}$

将后面两个式子代入第一个式子得 $\dfrac{\partial u}{\partial x} = f_x + f_y \varphi_x + f_y \varphi_t \psi_x$

方程组$\begin{cases} u = f(x,y,z) \\ y = \varphi(x,t) \\ t = \psi(x,z) \end{cases}$ 两边对z求导得 $\begin{cases} \dfrac{\partial u}{\partial z} = f_y \dfrac{\partial y}{\partial z} + f_z \\ \dfrac{\partial y}{\partial z} = \varphi_t \dfrac{\partial t}{\partial z} \\ \dfrac{\partial t}{\partial z} = \psi_z \end{cases}$

将后面两个式子代入第一个式子得 $\dfrac{\partial u}{\partial z} = f_y \varphi_t \psi_z + f_z$

所以 $\dfrac{\partial u}{\partial x} = f_x + f_y \varphi_x + f_y \varphi_t \psi_x$

$$\frac{\partial u}{\partial z} = f_y \varphi_t \psi_z + f_z$$

思考　　设 $v=f(y-z,z-x,x-y)$　试证:

$$v_x+v_y+v_z=0$$

解　设　$u=y-z,\ t=z-x,\ \omega=x-y$　，　则

$$v=f(u,t,\omega),\ u=y-z,\ t=z-x,\ \omega=x-y$$

利用链式法则有

$$v_x=f_1\frac{\partial u}{\partial x}+f_2\frac{\partial t}{\partial x}+f_3\frac{\partial \omega}{\partial x}=-f_2+f_3$$

$$v_y=f_1\frac{\partial u}{\partial y}+f_2\frac{\partial t}{\partial y}+f_3\frac{\partial \omega}{\partial y}=f_1-f_3$$

$$v_z=f_1\frac{\partial u}{\partial z}+f_2\frac{\partial t}{\partial z}+f_3\frac{\partial \omega}{\partial z}=-f_1+f_2$$

$$\Rightarrow\ v_x+v_y+v_z=-f_2+f_3+f_1-f_3-f_1+f_2=0$$

二、隐函数微分法

一. 一个方程的情形

(1) 由方程 $F(x,y)=0$ 确定的隐函数 $y=y(x)$,

求 $\dfrac{\mathrm{d}y}{\mathrm{d}x}$.

由于 $F[x,y(x)]=0$, 两边对 x 求导:

$$\frac{\partial F}{\partial x}+\frac{\partial F}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}x}=0$$

若 $\dfrac{\partial F}{\partial y}\neq 0$, 则有 $\dfrac{\mathrm{d}y}{\mathrm{d}x}=-\dfrac{F_x}{F_y}$

利用微分形式不变性

对 $F(x,y)=0$ 两边求微分：

$$F_x dx + F_y dy = 0$$

$$\frac{dy}{dx} = -\frac{F_x}{F_y}$$

(2) 方程的变量多于两个的 情形.

由方程 $F(x,y,z)=0$ 确定 $z=z(x,y)$，求 $\dfrac{\partial z}{\partial x}, \dfrac{\partial z}{\partial y}$.

由于 $F[x,y,z(x,y)]=0$， 两边对 x 求导：

$$\frac{\partial F}{\partial x}+\frac{\partial F}{\partial z}\frac{\partial z}{\partial x}=0$$

若 $\dfrac{\partial F}{\partial z}\neq 0$， 则有 $\dfrac{\partial z}{\partial x}=-\dfrac{F_x}{F_z}$

两边对 y 求导： $\dfrac{\partial F}{\partial y}+\dfrac{\partial F}{\partial z}\dfrac{\partial z}{\partial y}=0$

若 $\dfrac{\partial F}{\partial z}\neq 0$， 则有 $\dfrac{\partial z}{\partial y}=-\dfrac{F_y}{F_z}$

利用微分形式不变性

对 $F(x,y,z)=0$ 两边求微分：

$$F_x dx + F_y dy + F_z dz = 0$$

$$dz = -\frac{F_x dx + F_y dy}{F_z} = \frac{-F_x}{F_z}dx + \frac{-F_y}{F_z}dy$$

所以 $z_x = \dfrac{-F_x}{F_z}, \quad z_y = \dfrac{-F_y}{F_z}$

推广 由方程 $F(x_1,x_2,\cdots,x_n,z)=0$ 确定 $z=z(x_1,x_2,\cdots,x_n)$,

则 $\dfrac{\partial z}{\partial x_i} = -\dfrac{F_{x_i}}{F_z} \qquad (i=1,2,\cdots,n).$

二. 方程组的情形

用微分形式不变性也可 类似推出！

(1) 设 $\begin{cases} F(x,y,z)=0 \\ G(x,y,z)=0 \end{cases}$ 在满足一定条件下，可将 y, z 确定为

x 的函数，即 $y=y(x)$，$z=z(x)$，求 $\dfrac{\mathrm{d}y}{\mathrm{d}x}$，$\dfrac{\mathrm{d}z}{\mathrm{d}x}$.

$\because \quad F[x,y(x),z(x)]=0, \quad G[x,y(x),z(x)]=0,$

以上两式同时对 x 求导，得

$$F_x+F_y\cdot\frac{\mathrm{d}y}{\mathrm{d}x}+F_z\frac{\mathrm{d}z}{\mathrm{d}x}=0 \qquad G_x+G_y\cdot\frac{\mathrm{d}y}{\mathrm{d}x}+G_z\frac{\mathrm{d}z}{\mathrm{d}x}=0$$

解得：

$$\frac{\mathrm{d}y}{\mathrm{d}x}=-\frac{\begin{vmatrix} F_x & F_z \\ G_x & G_z \end{vmatrix}}{\begin{vmatrix} F_y & F_z \\ G_y & G_z \end{vmatrix}},\qquad \frac{\mathrm{d}z}{\mathrm{d}x}=-\frac{\begin{vmatrix} F_y & F_x \\ G_y & G_x \end{vmatrix}}{\begin{vmatrix} F_y & F_z \\ G_y & G_z \end{vmatrix}},$$

例　设 $x = u\cos\frac{v}{u}$, $y = u\sin\frac{v}{u}$, 求 $\frac{\partial u}{\partial x}$, $\frac{\partial v}{\partial x}$.

解　将 u, v 视为 x, y 的函数，方程两边同时对 x 求导，

有
$$\begin{cases} 1 = \frac{\partial u}{\partial x}\cdot\cos\frac{v}{u} - u\sin\frac{v}{u}\cdot\frac{u\frac{\partial v}{\partial x} - v\frac{\partial u}{\partial x}}{u^2} \\ 0 = \frac{\partial u}{\partial x}\cdot\sin\frac{v}{u} + u\cos\frac{v}{u}\cdot\frac{u\frac{\partial v}{\partial x} - v\frac{\partial u}{\partial x}}{u^2} \end{cases}$$

解得：
$$\frac{\partial u}{\partial x} = \cos\frac{v}{u},$$

$$\frac{\partial v}{\partial x} = -\sin\frac{v}{u} + \frac{v}{u}\cos\frac{v}{u}.$$

例　设函数 $u(x)$ 是由方程组

$$\begin{cases} u = f(x,y) & (7) \\ g(x,y,z) = 0 & (8) \\ h(x,z) = 0 & (9) \end{cases}$$

所确定，且 $\dfrac{\partial h}{\partial z} \neq 0$，$\dfrac{\partial g}{\partial y} \neq 0$，求 $\dfrac{\mathrm{d}u}{\mathrm{d}x}$.

解　由方程组可知：y, z, u都是x的一元函数，

对每个方程关于 x求全导数，得

$$\frac{du}{dx}=\frac{\partial f}{\partial x}+\frac{\partial f}{\partial y}\frac{dy}{dx}$$

$$\frac{\partial g}{\partial x}+\frac{\partial g}{\partial y}\cdot\frac{\mathrm{d}y}{\mathrm{d}x}+\frac{\partial g}{\partial z}\cdot\frac{\mathrm{d}z}{\mathrm{d}x}=0$$

$$\frac{\partial h}{\partial x}+\frac{\partial h}{\partial z}\cdot\frac{\mathrm{d}z}{\mathrm{d}x}=0$$

$$\begin{cases} u=f(x,y) \\ g(x,y,z)=0 \\ h(x,z)=0 \end{cases}$$

解之得，

$$\frac{\mathrm{d}u}{\mathrm{d}x}=\frac{\partial f}{\partial x}+\frac{\partial f}{\partial y}\cdot\left[\left(\frac{\partial g}{\partial z}\frac{\partial h}{\partial x}-\frac{\partial g}{\partial x}\frac{\partial h}{\partial z}\right)\Big/\left(\frac{\partial g}{\partial y}\frac{\partial h}{\partial z}\right)\right].$$

$$\begin{cases} u = f(x,y) \\ g(x,y,z) = 0 \\ h(x,z) = 0 \end{cases}$$

解法二： $du = f_x dx + f_y dy$

$$g_x dx + g_y dy + g_z dz = 0$$

$$h_x dx + h_z dz = 0 \qquad dz = -\frac{h_x}{h_z} dx$$

$$dy = -\frac{g_x dx}{g_y} - \frac{g_z dz}{g_y}$$

$$= -\frac{g_x dx}{g_y} - \frac{g_z}{g_y}\left(-\frac{h_x}{h_z} dx\right) = \left(\frac{g_z}{g_y}\frac{h_x}{h_z} - \frac{g_x}{g_y}\right)dx$$

$$du = f_x dx + f_y\left(\frac{g_z}{g_y}\frac{h_x}{h_z} - \frac{g_x}{g_y}\right)dx$$

因此有 $\dfrac{du}{dx} = f_x + f_y\left(\dfrac{g_z}{g_y}\dfrac{h_x}{h_z} - \dfrac{g_x}{g_y}\right).$

例　设函数 $z=z(x,y)$ 由方程 $F(x+\frac{z}{y},y+\frac{z}{x})=0$ 给出，

则有等式 $x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=z-xy.$

证法一．设 $u=x+\frac{z}{y}$，$v=y+\frac{z}{x}$，

则有 $F(u,v)=0$ 两边对 x 求导，得

$$F_u(1+\frac{1}{y}\frac{\partial z}{\partial x})+F_v(-\frac{z}{x^2}+\frac{1}{x}\frac{\partial z}{\partial x})=0$$

$$\Rightarrow \frac{\partial z}{\partial x}=\frac{\frac{z}{x^2}F_v-F_u}{\frac{1}{y}F_u+\frac{1}{x}F_v}$$

函数 $z=z(x,y)$ 由方程 $F(x+\frac{z}{y},y+\frac{z}{x})=0$ 给出

将 $F(u,v)=0$ 两边对 y 求导，得

$$F_u(-\frac{z}{y^2}+\frac{1}{y}\frac{\partial z}{\partial y})+F_v(1+\frac{1}{x}\frac{\partial z}{\partial y})=0$$

$$\Rightarrow \frac{\partial z}{\partial y}=\frac{\frac{z}{y^2}F_u-F_v}{\frac{1}{y}F_u+\frac{1}{x}F_v}$$

$$\therefore \quad x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=\frac{x(\frac{z}{x^2}F_v-F_u)}{\frac{1}{y}F_u+\frac{1}{x}F_v}+\frac{y(\frac{z}{y^2}F_u-F_v)}{\frac{1}{y}F_u+\frac{1}{x}F_v}$$

$$=z-xy.$$

函数 $z=z(x,y)$ 由方程 $F(x+\frac{z}{y},y+\frac{z}{x})=0$ 给出

证法二 (利用微分形式的不变性)

$$F_1\cdot d(x+\frac{z}{y})+F_2\cdot d(y+\frac{z}{x})=0$$

即 $$F_1\cdot\left(dx+\frac{ydz-zdy}{y^2}\right)+F_2\cdot\left(dy+\frac{xdz-zdx}{x^2}\right)=0$$

于是 $$dz=-\frac{F_1-F_2\frac{z}{x^2}}{F_1\frac{1}{y}+F_2\frac{1}{x}}dx-\frac{-F_1\frac{z}{y^2}+F_2}{F_1\frac{1}{y}+F_2\frac{1}{x}}dy$$

所以 $$\frac{\partial z}{\partial x}=-\frac{F_1+F_2(-\frac{z}{x^2})}{F_1\frac{1}{y}+F_2\frac{1}{x}}$$

$$\frac{\partial z}{\partial y}=-\frac{F_1(-\frac{z}{y^2})+F_2}{F_1\frac{1}{y}+F_2\frac{1}{x}}$$ 因此有 $$x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=z-xy.$$

解法三：视为方程组 $\begin{cases} u = f(x,y,z), \\ y = \varphi(x,t), \\ t = \psi(x,z). \end{cases}$ 确定的隐函数

$u = u(x,z), y = y(x,z), t = t(x,z)$ 的偏导数问题.

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将后面两个式子代入第一个式子得 $\dfrac{\partial u}{\partial x} = f_x + f_y \varphi_x + f_y \varphi_t \psi_x$

方程组$\begin{cases} u = f(x,y,z) \\ y = \varphi(x,t) \\ t = \psi(x,z) \end{cases}$ 两边对z求导得 $\begin{cases} \dfrac{\partial u}{\partial z} = f_y \dfrac{\partial y}{\partial z} + f_z \\ \dfrac{\partial y}{\partial z} = \varphi_t \dfrac{\partial t}{\partial z} \\ \dfrac{\partial t}{\partial z} = \psi_z \end{cases}$

将后面两个式子代入第一个式子得 $\dfrac{\partial u}{\partial z} = f_y \varphi_t \psi_z + f_z$

所以 $\dfrac{\partial u}{\partial x} = f_x + f_y \varphi_x + f_y \varphi_t \psi_x$

$$\frac{\partial u}{\partial z} = f_y \varphi_t \psi_z + f_z$$